MOVZZ — Master Roadmap

> Reliability-Orchestrated Mobility Platform · Chennai, India Last updated: March 2026

Legend

Symbol	Meaning
☑ DONE	Implemented and working
🔄 IN PROGRESS	Partially implemented
☐ TODO	Not yet started
🤖 AI	New AI engine task (from AI Implementation Plan)

Grand Summary

Category	Total	Done	Remaining
Section 1 — Foundation	33	33	0
Section 2 — Platform Features	22	5	17
Section 3 — AI Intelligence Layer	23	0	23
TOTAL	**78**	**38**	**40**

Progress: 38/78 tasks (49%)

SECTION 1 — Foundation (33/33 Complete ☑)

Phase 1 — Core Backend (12/12 ☑)

#	Task	Status	Key File(s)
1	Express + TypeScript + Prisma scaffold	☑ DONE	`backend/src/index.ts`
2	PostgreSQL schema — Users, Providers, Bookings, BookingAttempts, BookingLogs	☑ DONE	`backend/prisma/schema.prisma`
3	Redis cache with in-memory fallback	☑ DONE	`backend/src/config/redis.ts`
4	OTP authentication (phone) with JWT	☑ DONE	`backend/src/controllers/auth.controller.ts`

#	Task	Status	Key File(s)
5	Fare engine — mode-specific rates, surge, Haversine×1.35, airport detection, paise precision	☑ DONE	`backend/src/services/fare.service.ts`
6	Provider scoring — basic reliability from historical success rate	☑ DONE	`backend/src/services/provider-scoring.service.ts`
7	Booking state machine — SEARCHING → CONFIRMED → IN_PROGRESS → COMPLETED / FAILED / CANCELLED / MANUAL_ESCALATION	☑ DONE	`backend/src/services/booking.service.ts`
8	Recovery service — 3-level retry, auto-escalation, ₹100 compensation credit	☑ DONE	`backend/src/services/recovery.service.ts`
9	Quotes API — ranked options with reliability rationale	☑ DONE	`backend/src/services/quotes.service.ts`
10	Booking API — create, poll status, cancel	☑ DONE	`backend/src/controllers/booking.controller.ts`
11	Admin dashboard API — stats, provider list	☑ DONE	`backend/src/controllers/admin.controller.ts`
12	Rate limiting, CORS, Helmet security headers	☑ DONE	`backend/src/index.ts`

Phase 2 — Frontend Prototype (7/7 ☑)

#	Task	Status	Key File(s)
13	React + Vite scaffold with Zustand state management	☑ DONE	`frontend/src/App.jsx`
14	5-screen flow: Landing → Auth → Transport → Destination → Results	☑ DONE	`frontend/src/App.jsx`
15	Transport mode selector — CAB (Economy/Comfort/Premium), Bike, Auto, Metro	☑ DONE	`frontend/src/App.jsx`
16	Results screen — ranked quote cards with reliability score, ETA, price, tags (BEST / CHEAPEST / PREMIUM)	☑ DONE	`frontend/src/App.jsx`

#	Task	Status	Key File(s)
17	Auth store — OTP send/verify, JWT storage, auth state	☑ DONE	`frontend/src/stores/authStore.ts`
18	Booking store — fetchQuotes, createBooking, Socket.IO state sync	☑ DONE	`frontend/src/stores/bookingStore.ts`
19	API client (axios) with JWT header injection	☑ DONE	`frontend/src/api/client.ts`

Phase 3 — Production Hardening (5/5 ☑)

#	Task	Status	Notes
20	WebSockets (Socket.IO) — `booking:state_changed` server-push, per-user rooms, JWT on handshake	☑ DONE	`backend/src/config/socket.ts`, `frontend/src/stores/bookingStore.ts`
21	Google OAuth2 — direct REST flow, no Passport.js, `phone = "oauth_google_<sub>"`	☑ DONE	`backend/src/controllers/oauth.controller.ts`
22	Mapbox geocoding autocomplete — 300ms debounce, India filter, `onMouseDown` before blur	☑ DONE	`frontend/src/App.jsx`
23	Interactive map — react-map-gl, green (pickup) + orange (dropoff) markers, auto-pan on coord change	☑ DONE	`frontend/src/App.jsx`
24	Chennai preset location chips with real lat/lng coordinates	☑ DONE	`frontend/src/App.jsx`

Phase 4 — Background Jobs / BullMQ (4/4 ☑)

#	Task	Status	Notes
25	BullMQ queue singletons — booking-timeout, recovery-retry, sms-dispatch	☑ DONE	`backend/src/config/queues.ts`
26	Booking timeout worker — auto-cancel SEARCHING after 5 min, issue compensation	☑ DONE	`backend/src/workers/booking-timeout.worker.ts`
27	Recovery retry worker — async provider retry, 2s delay, delegates to 3-level retry	☑ DONE	`backend/src/workers/recovery.worker.ts`

#	Task	Status	Notes
28	SMS dispatch worker — 3 retries, exponential backoff, Twilio-ready (mock in dev)	☑ DONE	`backend/src/workers/sms.worker.ts`

Phase 9 — Admin Panel (5/5 ☑)

#	Task	Status	Notes
29	Full Admin Panel UI at `/admin` — 4 tabs: Dashboard, Escalations, Providers, Metrics	☑ DONE	`frontend/src/Admin.jsx`
30	Live booking map — react-map-gl, color-coded markers by state, 10s poll + Socket.IO push	☑ DONE	`frontend/src/Admin.jsx`, `backend/src/controllers/admin.controller.ts`
31	Provider management — list, pause/resume, add new provider form	☑ DONE	`frontend/src/Admin.jsx`
32	Manual escalation queue — ops confirms MANUAL_ESCALATION bookings with provider ID	☑ DONE	`frontend/src/Admin.jsx`
33	Analytics tab — today's booking states, weekly revenue, top providers	☑ DONE	`frontend/src/Admin.jsx`

Phase 7 — Infrastructure (3/4 ☑)

#	Task	Status	Notes
34	Sentry error tracking — `@sentry/node` + `@sentry/react`, real DSN active, gated on env var	☑ DONE	`backend/src/index.ts`, `frontend/src/main.jsx`
35	Neon.tech — cloud Postgres DATABASE_URL swap ready in `backend/.env`	☑ DONE	`backend/.env` (commented format)
36	AWS S3 — presigned URL pattern, profile photo upload endpoint, avatar in app header	☑ DONE	`backend/src/services/s3.service.ts`, `backend/src/controllers/upload.controller.ts`

Phase 6 — Notifications (1/4 ☑)

#	Task	Status	Notes
37	Transactional email via Resend — confirmation, cancellation, compensation; fire-and-forget on state transitions	☑ DONE	`backend/src/services/email.service.ts`, `RESEND_API_KEY` set

Phase 10 — Mobile (1/6 ☑)

#	Task	Status	Notes
38	PWA — vite-plugin-pwa, service worker, web manifest, offline cache, installable on Android/iOS	☑ DONE	`frontend/vite.config.js`, `vite-plugin-pwa`

SECTION 2 — Platform Features (5/22 Done, 17 Remaining)

Phase 5 — Payments ☐ (0/4)

#	Task	Status	Notes
39	Razorpay — create order endpoint	☐ TODO	`POST /api/v1/payments/create-order` · needs `RAZORPAY_KEY_ID` + `RAZORPAY_KEY_SECRET`
40	Razorpay — verify payment + HMAC-SHA256 signature check	☐ TODO	`POST /api/v1/payments/verify` · transitions booking to CONFIRMED on success
41	Frontend — Razorpay Web Checkout modal	☐ TODO	CDN script, no new npm · opens after `createBooking()` succeeds
42	Provider payout tracking (T+2 terms)	☐ TODO	New `ProviderPayout` Prisma table

Phase 6 — Notifications (remaining) 🔄 (1/4 done)

#	Task	Status	Notes
43	Twilio SMS — real OTP delivery	☐ TODO	Replace `console.log` mock in `sms.worker.ts` · needs `TWILIO_ACCOUNT_SID`, `TWILIO_AUTH_TOKEN`, `TWILIO_PHONE_NUMBER`
44	FCM push notifications — booking state changes, driver ETA	☐ TODO	Needs React Native or PWA service worker
45	WhatsApp API via Twilio	☐ TODO	Same Twilio credentials, different endpoint

Phase 7 — Infrastructure (remaining) 🔄 (3/4 done)

#	Task	Status	Notes
46	CDN — CloudFront or Cloudflare in front of S3	☐ TODO	Static asset delivery for frontend + S3 files

Phase 8 — Provider Integrations ☐ (0/4)

#	Task	Status	Notes
47	Fast Track Cabs API integration	☐ TODO	Real provider dispatch instead of mock scoring
48	Chennai Call Taxi API integration	☐ TODO	Local fleet operator
49	Uber/Ola aggregator fallback	☐ TODO	When all MOVZZ providers fail — last resort layer
50	Provider onboarding portal — KYC + doc upload	☐ TODO	S3 `providerDocKey()` already wired, needs admin UI + flow

Phase 10 — Mobile Apps (remaining) 🔄 (1/6 done)

#	Task	Status	Notes
51	React Native user app — port 5-screen flow	☐ TODO	Expo recommended · reuse existing API + WebSocket
52	React Native driver app — accept/reject rides, navigation	☐ TODO	Separate app with driver-specific Socket.IO events
53	Driver location tracking — real-time GPS updates	☐ TODO	New `driver:location_update` Socket.IO event
54	In-app FCM push notifications	☐ TODO	Firebase + React Native Push Notification library
55	App Store + Play Store deployment	☐ TODO	Expo EAS build

SECTION 3 — AI Intelligence Layer 🤖 (0/23 — All New)

> **Current state:** Provider scoring uses a basic historical success rate. No context, no orchestration strategy, no demand forecasting, no dynamic pricing.
>
> This section implements 7 AI engines across 4 weeks to transform MOVZZ into a true reliability-orchestrated platform.
>
> **Target outcomes:** Booking success rate 75% → 92%+, confirmation time 45s → <20s, revenue +10–15% via fair pricing.

Week 1 — Core AI Foundation (0/8) ☐

#	Task	Status	Files
AI-1	TypeScript AI interfaces	☐ TODO	`backend/src/types/ai.types.ts` NEW · interfaces: `RideContext`, `ProviderMetrics`, `ReliabilityScore`, `ScoreBreakdown` · enums: `Zone`, `WeatherCondition`, `TrafficLevel`, `RideType`
AI-2	Context Builder Service	☐ TODO	`backend/src/services/context-builder.service.ts` NEW · `buildRideContext(bookingData)` · extracts zone from lat/lng, Haversine distance, time factors (hour/day/peak/weekend), ride type (airport/corporate/personal), weather + traffic (mock OK for Week 1)
AI-3	Provider Metrics Service	☐ TODO	`backend/src/services/provider-metrics.service.ts` NEW · `getProviderMetrics(providerId)` · aggregates last 1000 bookings by hour (0–23) / day (0–6) / zone / distance range · recent perf (1h/6h/24h/7d) · consecutive streaks · Redis cache 5-min TTL
AI-4	Reliability Predictor (rewrite scoring)	☐ TODO	`backend/src/services/provider-scoring.service.ts` REWRITE · `predictReliability(providerId, context)` → 0–100 score · 8 contextual adjustments: time (±10), zone (±15), weather (±8), distance (±5), capacity (±12), recent perf (±10), ride type (±5), user match (±3) · outputs: score, confidence, reasoning[], warnings[]
AI-5	Orchestration Service	☐ TODO	`backend/src/services/ai/orchestration.service.ts` NEW · 5 strategies: SEQUENTIAL (top score >95%), PARALLEL_2 (>85%), PARALLEL_3 (>70%), CASCADE (>50%), EMERGENCY (<50%) · `Promise.race` for parallel · cancels non-winning providers · logs all BookingAttempts
AI-6	Failure Detector Service	☐ TODO	`backend/src/services/ai/failure-detector.service.ts` NEW · risk score: time-in-SEARCHING>60s (+20), rejections>2 (+30), all-high-reliability-rejected (+40), supply<0.5 (+25) · interventions: 30% queue backups / 50% expand radius / 70% offer driver bonus / 85% emergency mode / 90% auto-compensate
AI-7	Prisma migration — AI fields	☐ TODO	`backend/prisma/schema.prisma` · add to Booking: `orchestrationStrategy`, `orchestrationReasoning`, `aiReliabilityScore Float?`, `contextSnapshot Json?`, `timeToConfirm Int?` · add to Provider: `currentActiveRides Int`, `maxCapacity Int`, `last24hSuccessRate Float?`, `last7dSuccessRate Float?` · new model: `BookingAttempt` (id, bookingId, providerId, attemptNumber, status, responseTime, timestamp)
AI-8	Wire AI into booking.service.ts	☐ TODO	`backend/src/services/booking.service.ts` MODIFY · `createBooking()` flow: `buildRideContext` → score all providers → `decideStrategy` → `executeStrategy` → log BookingAttempts in DB · store `orchestrationStrategy` + `aiReliabilityScore` on Booking record

Week 2 — Performance + Visible AI in UI (0/5) ☐

#	Task	Status	Files
AI-9	Nightly Aggregation Job	☐ TODO	`backend/src/jobs/nightly-aggregation.job.ts` NEW · BullMQ cron at 2 AM · per-provider: success rates by hour/day/zone/distance range, last 24h/7d rates, preferred zones · stores in `ProviderMetricsCache` table · Redis also updated
AI-10	Cache Service	☐ TODO	`backend/src/services/cache.service.ts` NEW · `getOrCompute(key, ttl, fn)` pattern · key prefixes: `provider:{id}:metrics` (5min), `context:weather:{lat},{lng}` (30min), `context:traffic` (5min), `zone:mapping` (1day)
AI-11	ML Data Collection Worker	☐ TODO	`backend/src/workers/ml-data-collection.worker.ts` NEW · triggers after COMPLETED/FAILED · logs: full RideContext, all provider scores at time of booking, selected provider, orchestration strategy, actual outcome, `timeToConfirm`, `timeToComplete` · stored in `MLTrainingData` table
AI-12	Prisma migration — cache + ML tables	☐ TODO	`backend/prisma/schema.prisma` · new: `ProviderMetricsCache` (providerId, metricType, metricKey, metricValue, computedAt, unique constraint) · new: `MLTrainingData` (contextSnapshot Json, providerScores Json, selectedProviderId, orchestrationStrategy, actualOutcome, timeToConfirm)
AI-13	Frontend — AI-enhanced Results screen	☐ TODO	`frontend/src/App.jsx` MODIFY · reliability score progress bar (green >90 / yellow >75 / red <75) · AI reasoning text below each quote card · tags as colored badges (MOST_RELIABLE / CHEAPEST / PREMIUM) · new Socket.IO events: `booking:ai_decision`, `booking:provider_queried`, `booking:failure_risk_update` · live status text: "Analyzing providers..." → "Querying Fast Track..." → "Confirmed!"

Week 3 — Demand Forecasting + Fair Dynamic Pricing (0/5) ☐

#	Task	Status	Files
AI-14	Demand Forecaster Service	☐ TODO	`backend/src/services/ai/demand-forecaster.service.ts` NEW · pattern matching (no ML yet) · per zone/hour: avg rides last 4 weeks same day/hour, trend (increasing/decreasing), day-of-week multiplier, event multiplier (manual CSV of Chennai events: concerts, cricket, holidays), weather multiplier (rain +30%) · outputs: predictedRides + confidenceLow/High interval
AI-15	Prisma migration — demand forecasts	☐ TODO	`backend/prisma/schema.prisma` · new: `DemandForecast` (zone, forecastHour DateTime, predictedRides Float, confidenceLow Float, confidenceHigh Float, actualRides Int?, forecastAccuracy Float?, unique on zone+forecastHour)

#	Task	Status	Files
AI-16	Proactive driver positioning job	☐ TODO	`backend/src/jobs/demand-forecast-update.job.ts` NEW · nightly: generate 24h forecasts, detect supply gaps (predictedRides > available drivers in zone) · trigger: WhatsApp/SMS to drivers in shortage zone ("High demand 6–9 AM Airport. ₹100 bonus per ride") · track effectiveness in DB
AI-17	Enhanced fare.service.ts — dynamic pricing	☐ TODO	`backend/src/services/fare.service.ts` ENHANCE · add demand multiplier (+5–15%, demand > supply) · weather: clear 0% / rain +8% / heavy rain +15% / storm +20% · traffic: low –5% / high +8% / jam +12% · time-of-day: late-night +10% / peak +5% · **hard cap: base fare × 1.2 max (never more than +20%)** · Movzz Pass users always get base fare (no surge)
AI-18	Pricing breakdown API + frontend	☐ TODO	`backend/src/controllers/booking.controller.ts` MODIFY · quotes response adds: `{ fareEstimate, baseFare, breakdown: [{factor, amount}], explanation }` · `frontend/src/App.jsx`: tap quote card → show pricing breakdown modal with "Why this price?" explanation

Week 4 — Provider Analytics + User Personalization (0/5) ☐

#	Task	Status	Files
AI-19	Provider Analytics Service	☐ TODO	`backend/src/services/ai/provider-analytics.service.ts` NEW · mine acceptance patterns (which zones/distances/fares accepted most) · behavioral profiles: airport specialist, short-ride vs long-distance, morning person vs night owl, price-sensitive vs volume-focused · outputs: human-readable recommendations ("Send them more airport rides 6–9 AM")
AI-20	Provider Dashboard Endpoint	☐ TODO	`backend/src/controllers/provider-dashboard.controller.ts` NEW · `GET /api/v1/providers/:id/insights` · response: `overallStats`, `yourBestTimes[]`, `yourPreferredZones[]`, `earningsOptimization { recommendation, potential }`, `aiInsights[]`
AI-21	User Personalization Service	☐ TODO	`backend/src/services/ai/personalization.service.ts` NEW · `buildUserProfile(userId)`: favorite providers (repeat 5-star), commute detection (same route weekly), pre-book vs last-minute, price sensitivity · `GET /api/v1/users/me/recommendations`: pre-filled route, suggested provider, Movzz Pass upsell if 12+ rides/month
AI-22	Prisma migration — personalization tables	☐ TODO	`backend/prisma/schema.prisma` · new: `UserPreference` (userId, preferenceType, preferenceKey, preferenceValue, confidence Float, learnedAt, lastUsedAt, unique on userId+type+key) · new: `UserPattern` (userId, patternType, patternData Json, confidence Float, detectedAt, lastOccurrence)

#	Task	Status	Files
AI-23	Admin AI Dashboard	☐ TODO	`backend/src/controllers/admin.controller.ts` ENHANCE + `frontend/src/Admin.jsx` MODIFY · new endpoints: `GET /admin/ai-performance` (orchestration strategy distribution, prediction accuracy, success rate with vs without AI) · `GET /admin/demand-insights` (forecast vs actual, supply shortage alerts, zone-by-hour heatmap) · `GET /admin/provider-rankings` (top 10 reliable, underperforming list) · Admin panel: new AI Performance tab with charts

SECTION 4 — Summary by Phase

Phase	Description	Total	Done	Status
1	Core Backend	12	12	☑ Complete
2	Frontend Prototype	7	7	☑ Complete
3	Production Hardening	5	5	☑ Complete
4	Background Jobs (BullMQ)	4	4	☑ Complete
5	Payments (Razorpay)	4	0	☐ Not started
6	Notifications	4	1	🔄 In progress
7	Infrastructure	4	3	🔄 In progress
8	Provider Integrations	4	0	☐ Not started
9	Admin Panel	5	5	☑ Complete
10	Mobile Apps	6	1	🔄 In progress
AI Week 1	Core AI (Reliability + Orchestration + Failure Detection)	8	0	☐ Not started
AI Week 2	Performance + ML Data + AI in UI	5	0	☐ Not started
AI Week 3	Demand Forecasting + Dynamic Pricing	5	0	☐ Not started
AI Week 4	Analytics + Personalization	5	0	☐ Not started
TOTAL		**78**	**38**	**49% done**

SECTION 5 — Recommended Priority Order

Priority	Task(s)	Effort	Rationale
1	Twilio SMS — real OTP (#43)	2 hrs	OTP is still `console.log` — no real user can sign up
2	Razorpay payments (#39–41)	1 day	Core revenue flow — nothing monetizes without it

Priority	Task(s)	Effort	Rationale
3	AI Week 1 — Core AI (#AI-1 to AI-8)	1 week	Biggest product differentiator; enables orchestration + failure prevention
4	AI Week 2 — Performance + UI (#AI-9 to AI-13)	1 week	Makes AI visible to users; scores and reasoning shown in results screen
5	CDN (#46)	1 hr	Quick win — CloudFront/Cloudflare in front of S3
6	AI Week 3 — Forecasting + Pricing (#AI-14 to AI-18)	1 week	Revenue optimization + proactive driver supply management
7	AI Week 4 — Analytics + Personalization (#AI-19 to AI-23)	1 week	Provider retention + user engagement
8	WhatsApp notifications (#45)	2 hrs	Ride alerts via WhatsApp (Twilio, same credentials)
9	Provider integrations (#47–50)	2 days	Real dispatch network beyond mock scoring
10	React Native (#51–55)	3–4 weeks	Full mobile product for both users and drivers

AI Success Metrics (Track Weekly After AI Week 1)

Metric	Baseline (Now)	Target (Post-AI)
Booking success rate	~75%	92%+
Avg time to confirmation	~45 seconds	<20 seconds
Rides needing manual ops	~20%	<5%
AI prediction accuracy	—	>90% (predicted vs actual outcome)
Revenue from dynamic pricing	0%	+10–15%
User satisfaction (avg rating)	~4.2	4.7+

*MOVZZ Master Roadmap · March 2026 · **38/78 tasks done (49%)***